

SECTION A

Q1. a) $\lim_{x \rightarrow 2} \frac{10 - 3x - x^2}{x - 2}$

$= \lim_{x \rightarrow 2} \frac{-(x^2 + 3x - 10)}{x - 2}$ — (K1) factorize correctly

$= \lim_{x \rightarrow 2} \frac{-(x + 5)(x - 2)}{x - 2}$

$= \lim_{x \rightarrow 2} -(x + 5)$ } — (K1) simplify & subst.

$= -(2 + 5)$

$= -7$ — (21)

b) $\lim_{x \rightarrow \infty} \frac{x + 2}{x - 1}$

$= \lim_{x \rightarrow \infty} \frac{\frac{x}{x} + \frac{2}{x}}{\frac{x}{x} - \frac{1}{x}}$ — (K1)

$= \lim_{x \rightarrow \infty} \frac{1 + \frac{2}{x}}{1 - \frac{1}{x}}$

$= \frac{1 + 0}{1 - 0}$ — (K1)

$= 1$ — (21)

c) $\lim_{x \rightarrow 3^-} \frac{|x - 3|}{x^2 - 2x - 3} = \begin{cases} +(x - 3) ; x \geq 3^+ \\ -(x - 3) ; x < 3^- \end{cases}$

$= \lim_{x \rightarrow 3^-} \frac{-(x - 3)}{(x - 3)(x + 1)}$ — (B1) correct numerator

$= \lim_{x \rightarrow 3^-} \frac{-(x - 3)}{(x - 3)(x + 1)}$ — (K1) factorize & simplify

$= -\frac{1}{3 + 1}$ — (K1) Subst.

$= -\frac{1}{4}$ — (21)

Total: 10 marks

Q2. a) $f'(2^-) = \lim_{x \rightarrow 2^-} \frac{x^2 - 3 - [(2)^2 - 3]}{x - 2}$ — (B1) calc f(2)

$= \lim_{x \rightarrow 2^-} \frac{x^2 - 3 - 1}{x - 2}$ — (K1) correct function

$= \lim_{x \rightarrow 2^-} \frac{(x - 2)(x + 2)}{x - 2}$

$= 2 + 2$

$= 4$

$f'(2^+) = \lim_{x \rightarrow 2^+} \frac{\sqrt{x - 1} - (\sqrt{2 - 1})}{x - 2}$

$= \lim_{x \rightarrow 2^+} \frac{\sqrt{x - 1} - 1}{x - 2}$

$= \lim_{x \rightarrow 2^+} \frac{\sqrt{x - 1} - 1}{x - 2} \times \frac{\sqrt{x - 1} + 1}{\sqrt{x - 1} + 1}$

$= \lim_{x \rightarrow 2^+} \frac{x - 2}{(x - 2)(\sqrt{x - 1} + 1)}$ — (K1) correct func. & x conjugate

$= \frac{1}{\sqrt{2 - 1} + 1}$

$= \frac{1}{2}$ — (21) both value correct

$f'(2^-) \neq f'(2^+) @ \lim_{x \rightarrow 2} f(x)$ does not exist
 $\therefore f(x)$ is not differentiable at $x = 2$

— (21) reason & conclusion

b) $y = (x^2 + 1)(\sqrt{1 - x^2})$

$\frac{dy}{dx} = (x^2 + 1) \left[\frac{1}{2}(1 - x^2)^{-\frac{1}{2}}(-2x) \right] + \sqrt{1 - x^2}(2x)$

— (K1) correct u.v either one correct

— (21) both correct

$= \frac{-x(x^2 + 1)}{\sqrt{1 - x^2}} + 2x\sqrt{1 - x^2}$

$= \frac{-x^3 - x + 2x - 2x^3}{\sqrt{1 - x^2}}$ } — (K1) equate denominator simplify this

$= \frac{3(1 - 3x^2)}{\sqrt{1 - x^2}}$ — (21) any equivalent form

Q3. $\frac{r}{h} = \frac{4}{16}$

$r = \frac{1}{4}h$ — (BI) r in terms of h

$V = \frac{\pi}{3}r^2h$

$V = \frac{\pi}{3}\left(\frac{1}{4}h\right)^2h$
 $= \frac{\pi}{48}h^3$ — (KI) subs. his r into correct formula

$\frac{dV}{dh} = \frac{\pi}{16}h^2$ — (KI)

$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$
 $= \frac{16}{\pi h^2} \times -2$ — (KI) apply chain rule
 $= \frac{-32}{\pi(8)^2}$ — (BI) $\frac{dV}{dt} = -2$
 $= -\frac{1}{2\pi} / -0.159$ — (AI)

SECTION B

Total: 6 marks

Q1. let $m = x + yi$
 $n = x - yi$

$-2m + (3+i)n = -4+3i$

$-2(x+yi) + (3+i)(x-yi) = -4+3i$ — (BI) subst.
 $-2x-2yi + 3x-3yi+xi-yi^2 = -4+3i$

$x+y+(x-5y)i = -4+3i$

$x+y = -4$
 $x-5y = 3$ — (KI)

$(-4-y)-5y = 3$

$-4-6y = 3$

$6y = -7$

$y = -\frac{7}{6}$

$x - \frac{7}{6} = -4$ — (KI) attempt to solve until get $x \neq y$
 $x = -\frac{17}{6}$

$\therefore m = -\frac{17}{6} - \frac{7}{6}i$ — (AI)

$n = -\frac{17}{6} + \frac{7}{6}i$ — (AI)

Q2. a) $\log_9(\log_{\sqrt{p}} \frac{1}{27}) = \frac{1}{2}$

$\log_{\sqrt{p}} \frac{1}{27} = 9^{\frac{1}{2}}$ — (BI) index form

$\frac{1}{27} = (\sqrt{p})^3$

$p^{\frac{3}{2}} = \frac{1}{27}$ — (BI) index form

$\therefore p = \left(\frac{1}{27}\right)^{\frac{2}{3}}$ — (KI) p as subject

$= \frac{1}{9}$ — (AI)

b) $2^x + 4 < 2^{2x} + 2^x < 12$

$2^x + 4 < 2^{2x} + 2^x$ — (BI) both LHS & RHS

$2^{2x} > 4$

let $2^x = y$

$y^2 - 4 > 0$

$(y-2)(y+2) > 0$ — (KI)



$y < -2$ or $y > 2$ and

$2^x > 2$ — (KI)
 $x > 1$

$2^{2x} + 2^x < 12$

$y^2 + y - 12 < 0$ — (KI)

$(y+4)(y-3) < 0$



$-4 < y < 3$

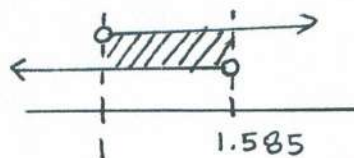
$2^x > -4$

$2^x < 3$ — (KI) both

no solution since $2^x > 0$ for all x .

$x < \frac{\log 3}{\log 2}$ — (AI)

$x < 1.585$



$\therefore \{x : 1 < x < 1.585\}$ — (AI)
 $(1, 1.585)$

Q3 a) $f(x) = \frac{2e^{\frac{3}{2}x}}{e^{3x}-1}$, $g(x) = \frac{e^{3x}+1}{e^{3x}-1}$

$$[f(x)]^2 - [g(x)]^2$$

$$= \left[\frac{2e^{\frac{3}{2}x}}{e^{3x}-1} \right]^2 - \left[\frac{e^{3x}+1}{e^{3x}-1} \right]^2$$

$$= \frac{4e^{3x}}{(e^{3x}-1)^2} - \frac{e^{6x}+2e^{3x}+1}{(e^{3x}-1)^2} \quad \left. \begin{array}{l} \text{---(B1) subst.} \\ \text{---(B1) expand} \end{array} \right\}$$

$$= \frac{-e^{6x}+2e^{3x}-1}{(e^{3x}-1)^2} \quad \text{---(K1) as it is}$$

$$= -\frac{(e^{3x}-1)(e^{3x}-1)}{(e^{3x}-1)^2} \quad \text{---(K1) factorize \& simplify}$$

$$= -1 \quad \text{---(A1)}$$

b) $g(x) = (x-2)^2 - 6$, $h(x) = 3x^2 - 12x + 2$

$$f[g(x)] = h(x)$$

$$f[(x-2)^2 - 6] = 3x^2 - 12x + 2$$

$$\text{let } y = (x-2)^2 - 6$$

$$x = \sqrt{y+6} + 2 \quad \text{---(K1) must correct}$$

$$f[y] = 3(\sqrt{y+6} + 2)^2 - 12(\sqrt{y+6} + 2) + 2$$

$$= 3(y+6+4\sqrt{y+6}+4) - 12\sqrt{y+6} - 24 + 2$$

$$= 3y+30+12\sqrt{y+6}-12\sqrt{y+6}-22$$

$$= 3y+8 \quad \text{---(K1) in terms of y linear form}$$

$$\therefore f(x) = 3x+8 \quad \text{---(A1)}$$

Total : 8 marks

Q4. $f(x) = 2 - \sqrt{3-x}$

(a) $3-x > 0$

$$-x > -3$$

$$x \leq 3 \quad D_f : (-\infty, 3] \quad \text{---(B1)}$$

$$y = 2 - \sqrt{3-3}$$

$$= 2 \quad R_f : (-\infty, 2] \quad \text{---(B1)}$$

(b) $f(x_1) = f(x_2)$

$$2 - \sqrt{3-x_1} = 2 - \sqrt{3-x_2} \quad \text{---(K1) subst.}$$

$$-\sqrt{3-x_1} = -\sqrt{3-x_2}$$

$$3-x_1 = 3-x_2$$

$$x_1 = x_2$$

since $x_1 = x_2 \therefore f(x)$ is 1-1 function

---(A1)

(c) $f[f^{-1}(x)] = x$

$$2 - \sqrt{3-f^{-1}(x)} = x \quad \text{---(B1) subst } f^{-1}$$

$$-\sqrt{3-f^{-1}(x)} = x-2$$

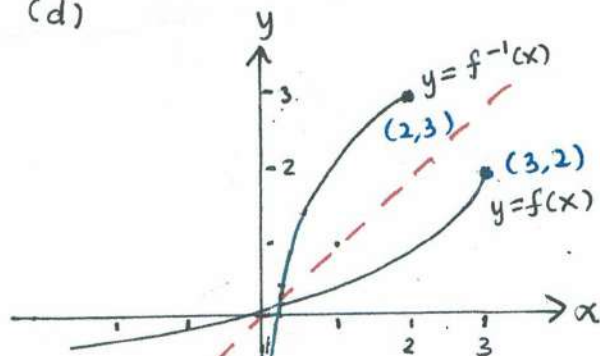
$$3-f^{-1}(x) = (2-x)^2 \quad \left. \begin{array}{l} \text{---(K1) attempt to find} \\ \text{---(K1) } f^{-1}(x) \end{array} \right\}$$

$$\therefore f^{-1} = 3 - (2-x)^2$$

---(A1) AEF

$$D_{f^{-1}} : (-\infty, 2] \quad \text{---(B1) must find } f^{-1} \text{ @ } \{x: x \leq 2\}$$

(d)



RI - $f(x)$ correct shape intercept y-axis

RI - $f^{-1}(x)$ correct shape intercept x-axis

RI - all correct + label

Q5. a) $\lim_{x \rightarrow a} \frac{\sqrt{x}-2}{x-4} = \frac{1}{6}$

$= \lim_{x \rightarrow a} \frac{\sqrt{x}-2}{x-4} \times \frac{\sqrt{x}+2}{\sqrt{x}+2}$

$= \lim_{x \rightarrow a} \frac{x-4}{(x-4)(\sqrt{x}+2)}$

$= \frac{1}{\sqrt{a}+2} = \frac{1}{6}$ — (B1)

$6 = \sqrt{a} + 2$

$\sqrt{a} = 4$

$\therefore a = 16$ — (J1)

(b)

$$f(x) = \begin{cases} 1 - 2e^{2x} & ; x < 0 \\ 1 & ; x = 0 \\ e^{3x} - 2 & ; x > 0 \end{cases}$$

(i). $\lim_{x \rightarrow 0^-} 1 - 2e^{2x}$ — (K1) choose correct function both

$= 1 - 2e^{2(0)}$

$= -1$

$\lim_{x \rightarrow 0^+} e^{3x} - 2$

$= e^{3(0)} - 2$

$= -1$

subst. $x = 0$

both get -1

since $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$

$\therefore \lim_{x \rightarrow 0} f(x) = -1$ / exist

conclusion

(ii) T1: $f(0) = 1$ — (K1)

T2: $\lim_{x \rightarrow 0} f(x) = -1$

T3: $f(0) \neq \lim_{x \rightarrow 0} f(x)$ — (J1)

$\therefore f(x)$ is not continuous at $x = 0$. — (J1)

Q5. c) $f(x) = \frac{\sqrt{x^2+5}}{x-2}$

V.A

$x-2=0$

$x=2$

$\lim_{x \rightarrow 2^-} \frac{\sqrt{x^2+5}}{x-2} = \frac{3}{0} = -\infty$

$\lim_{x \rightarrow 2^+} \frac{\sqrt{x^2+5}}{x-2} = \frac{3}{0} = +\infty$

$\therefore x=2$ is vertical asymptote. — (J1)

H.A

$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^2+5}}{x-2} = \frac{\infty}{\infty}$

$= \lim_{x \rightarrow +\infty} \frac{\sqrt{\frac{x^3}{x^2} + \frac{5}{x^2}}}{\frac{x}{x} - \frac{2}{x}}$

divide by x^2 (numerator) get $+\infty$

$= \lim_{x \rightarrow +\infty} \frac{\sqrt{x + \frac{5}{x^2}}}{1 - \frac{2}{x}}$

$= \frac{\sqrt{\infty+0}}{1-0}$

$= \infty$

$\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2+5}}{x-2}$

$= \lim_{x \rightarrow -\infty} \frac{-\sqrt{\frac{x^3}{x^2} + \frac{5}{x^2}}}{\frac{x}{x} - \frac{2}{x}}$ — (K1)

$= \lim_{x \rightarrow -\infty} \frac{-\sqrt{x + \frac{5}{x^2}}}{1 - \frac{2}{x}}$

$= \frac{-\sqrt{-\infty}}{1-0}$

$=$ does not exist.

$\therefore$ Horizontal asymptote does not exist. — (J1)

Q6. (a) $xy - \sin(xy) - 3 = 0$

(i) $x \frac{dy}{dx} + 1 \cdot y$ — (K1) diff all terms either one correct

$$-[\cos(xy) \cdot [x \frac{dy}{dx} + y]] = 0$$

$$x \frac{dy}{dx} - x \cos(xy) \frac{dy}{dx} = y \cos xy - y$$

— (A1) all correct

$$\frac{dy}{dx} (x - x \cos(xy)) = y (\cos xy - 1)$$

$$\frac{dy}{dx} = \frac{y (\cos xy - 1)}{x (1 - \cos xy)}$$

(K1) attempt to get $\frac{dy}{dx}$

$$= \frac{-y (1 - \cos xy)}{x (1 - \cos xy)}$$

$$= -\frac{y}{x} \text{ — (A1)}$$

(ii) $\frac{d^2y}{dx^2} = \frac{x(-\frac{dy}{dx}) - (-y)(1)}{x^2}$ — (K1)

$$= \frac{x(\frac{y}{x}) + y}{x^2}$$

attempt to find $\frac{d^2y}{dx^2}$

$$= \frac{2y}{x^2}$$

LHS:

$$x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx}$$

$$= x \left[\frac{2y}{x^2} \right] + 2 \left[-\frac{y}{x} \right]$$

$$= \frac{2y}{x} - \frac{2y}{x}$$

$$= 0 \text{ — (A1)}$$

b) $x(t) = \ln(t^2 + t)$

$$\frac{dx}{dt} = \frac{1}{t^2 + t} \cdot 2t + 1$$

$$= \frac{2t + 1}{t^2 + t}$$

$$y(t) = \sqrt{t^2 + t} = (t^2 + t)^{1/2}$$

$$\frac{dy}{dt} = \frac{1}{2} (t^2 + t)^{-1/2} (2t + 1)$$

$$= \frac{2t + 1}{2 \sqrt{t^2 + t}}$$

— (K1) find $\frac{dx}{dt}$ @ $\frac{dy}{dt}$

— (A1) both correct

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$= \frac{2t + 1}{2 \sqrt{t^2 + t}} \times \frac{(t^2 + t)}{2t + 1}$$

(K1) subst. into chain rule

$$= \frac{1}{2} (t^2 + t)^{1/2} \text{ — (A1)}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left[\frac{1}{2} (t^2 + t)^{1/2} \right] \times \frac{t^2 + t}{2t + 1} \text{ — (K1)}$$

$$= \frac{1}{4} (t^2 + t)^{-1/2} (2t + 1) \times \frac{t^2 + t}{2t + 1}$$

— (K1) diff.

$$= \frac{1}{4} (t^2 + t)^{-1/2} (t^2 + t)$$

$$= \frac{1}{4} (t^2 + t)^{1/2} \text{ — (A1)}$$

When $x = \ln 2$

$$\ln 2 = \ln(t^2 + t)$$

$$t^2 + t = 2$$

$$t^2 + t - 2 = 0$$

— (B1) find t

$$(t + 2)(t - 1) = 0$$

$$t = 1, t = -2 \text{ reject}$$

since $t > 0$

When $t = 1$

$$\frac{d^2y}{dx^2} = \frac{1}{4} (1^2 + 1)^{1/2} \text{ — (K1)}$$

$$= \frac{\sqrt{2}}{4} \text{ — (A1)}$$

Q7. a) $f(x) = \cos x + \sin x$

$f'(x) = -\sin x + \cos x$

$-\sin x + \cos x = 0$ — (k) diff $\neq 0$

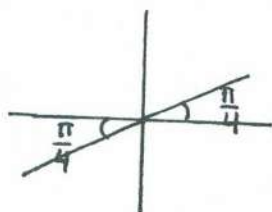
$\sin x = \cos x$

$\frac{\sin x}{\cos x} = 1$

$\tan x = 1$

$x = \frac{\pi}{4}$

must get $\tan x$



$\therefore x = \frac{\pi}{4}, \frac{5\pi}{4}$ — (a) both correct

int	$(0, \frac{\pi}{4})$	$(\frac{\pi}{4}, \frac{5\pi}{4})$	$(\frac{5\pi}{4}, 2\pi)$
x	$\frac{\pi}{8}$	π	$\frac{3\pi}{2}$
$f'(x)$	+	-	+
shape	/	\	/

— (k) table at least 1 x

$f(\frac{\pi}{4}) = \cos \frac{\pi}{4} + \sin \frac{\pi}{4} = \sqrt{2}$

$f(\frac{5\pi}{4}) = \cos \frac{5\pi}{4} + \sin \frac{5\pi}{4} = -\sqrt{2}$

(k) find y

$\therefore (\frac{\pi}{4}, \sqrt{2})$ is max point

$(\frac{5\pi}{4}, -\sqrt{2})$ is min point

(a)

b) $f'(x) = -\sin x + \cos x$

$f''(x) = -\cos x - \sin x$

$-\cos x - \sin x = 0$

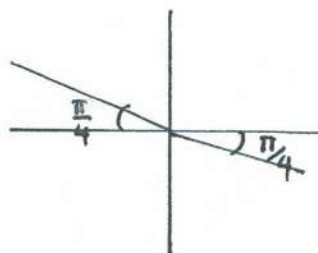
$-\sin x = \cos x$

$\frac{\sin x}{\cos x} = -1$

$\tan x = -1$

$x = \frac{\pi}{4}$

(k) diff $\neq 0$
Attempt to solve until get x.



$\therefore x = \frac{3\pi}{4}, \frac{7\pi}{4}$ — (a)

int	$(0, \frac{3\pi}{4})$	$(\frac{3\pi}{4}, \frac{7\pi}{4})$	$(\frac{7\pi}{4}, 2\pi)$
x	$\frac{3\pi}{8}$	$\frac{3\pi}{2}$	$\frac{15\pi}{8}$
$f'(x)$	-	+	-
Concave	∩	∪	∩

— (k)

$f(\frac{3\pi}{4}) = \cos(\frac{3\pi}{4}) + \sin(\frac{3\pi}{4}) = 0$

$f(\frac{7\pi}{4}) = \cos(\frac{7\pi}{4}) + \sin(\frac{7\pi}{4}) = 0$

$\therefore (\frac{3\pi}{4}, 0)$ & $(\frac{7\pi}{4}, 0)$

are points of inflection.

(a)

Total: 9 marks